

An Open Citation Graph of 38,357 Indian Supreme Court Judgments: Graph-Ranked Retrieval Lifts Small Models’ Hit Rate on Deciding Judgments from 30% to 83%

Shreyas Bile
[Department], [Institution]
[City], India
shreyasbile10@gmail.com

Abstract—Indian courts have begun to set aside decisions that relied on non-existent, AI-generated judgments. We build open infrastructure that checks Supreme Court citations and finds the judgments that decide a question, and we measure what it changes for language models. From the public AWS archive we build an audited index of 19,319,963 Indian judgments and orders, and a verifier for Supreme Court citations whose labels were correct in 96.8% of a held-out audit. From the text of all 38,357 reported Supreme Court judgments we extract the citations each makes and resolve them into an open citation graph of 109,370 links. Where a citation number allows a check, the links resolved by party names alone are correct in 98.4% of cases. A retriever over the reporters’ headnotes that ranks with the graph’s citation counts places 20.4% of the judgments that 1,000 held-out judgments cite in its top ten, 22% more than without the counts. For 33 of 40 legal questions its top three results include a judgment that decided the question. Four open-weight models answering from memory, with reasoning disabled, identified at most 1 of 202 real citations. They named a deciding judgment in 48 of 160 model-question cases, and only 40.0% of the cases they listed exist. Verifier feedback and search by party names removed most non-existent cases but found no better authorities. With search by subject and the citation prior, the same models named a deciding judgment in 133 of 160 cases, and 98.5% of the cases they listed exist. Existence checks catch fabrication; finding authority needs the text and the citation network.

Index Terms—legal information retrieval, citation network, hallucination, citation verification, large language models, retrieval-augmented generation, Indian case law, open data

I. INTRODUCTION

Large language models (LLMs) now draft legal research notes, pleadings and even adjudicatory orders, and they cite authorities that do not exist. In 2023 a United States federal court sanctioned lawyers who had filed a brief citing decisions fabricated by ChatGPT [1]. The same failure has now reached adjudicators in India. In a ruling reported in October 2025, the Bombay High Court set aside an income-tax assessment of INR 27.91 crore that relied on non-existent, AI-generated case law [2]. In February 2026 the Supreme Court of India declared that a trial-court decision resting on “non-existent and fake alleged judgments” is “not an error in the decision-making process” but misconduct [3], and in September 2026

it set aside a customs penalty of INR 425.27 crore that relied on non-existent judgments and on real ones that did not say what was attributed to them [4].

Two different checks are needed. The first is whether a cited judgment exists and the citation belongs to it. The second is whether it is an authority for the point: a judgment that decided the question, rather than one that merely exists. For United States case law an open API performs the first check against millions of citations [5], and the citation network of the Supreme Court has long been used to measure the importance of precedents [6]. For Indian case law the authoritative sources are the Supreme Court’s official law report, the Supreme Court Reports (S.C.R.), published digitally since 2024 [7], and a public archive of Supreme Court and High Court judgments on the AWS Open Data registry [8], [9]. Evaluations of LLMs in Indian legal practice report fabricated authorities and weak citation discipline [10], [11]. We are not aware of an open citation checker built on these sources, of an open citation graph of the Supreme Court (the largest open release of the archive’s text leaves it out of scope [12]), or of a citation-level measurement of how LLMs cite Indian judgments.

This paper provides all three. Our contributions are:

- **An audited open index and a citation verifier.** We load the metadata of the AWS archive into a PostgreSQL index of 19,319,963 judgments and orders and repair 86.9% of the 1,291,516 High Court records whose link to the source PDF is broken (§III). On the Supreme Court’s 38,366 reported judgments we build a verifier for neutral citations, S.C.R. citations and party names (§IV).
- **An open citation graph of the Supreme Court.** From the text of all 38,357 reported judgments we extract the citations each makes and resolve them to judgments in the index, giving 109,370 links. We measure their accuracy without labels, where citation numbers and case names can be compared, and by hand (§V).
- **Retrieval that finds authorities.** We search the reporters’ headnotes by subject and rank with the graph’s citation counts, and we evaluate on 1,000 held-out judgments, whose own citations serve as relevance labels, and on 40

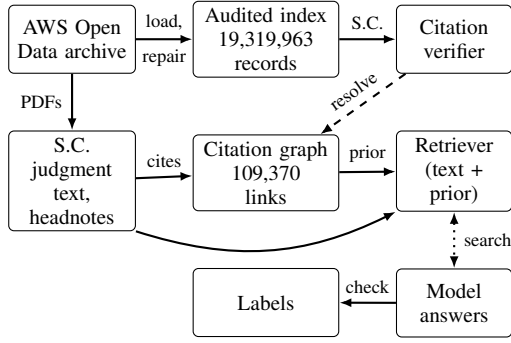


Fig. 1. Overview. The archive’s metadata is loaded into an audited index; the verifier resolves citations and case names against its Supreme Court part. The text of every Supreme Court judgment yields headnotes and the citations each judgment makes, which the verifier resolves into a citation graph. The retriever searches headnotes by subject and ranks with the graph’s citation counts. Model answers are checked by the verifier; in the search conditions the model queries the index or the retriever before answering.

legal questions (§VI).

- **A citation audit of open models under five conditions.** We evaluate four open-weight LLMs on 403 items each (§VII–§VIII). We compare answers from memory with answers after verifier feedback, with search by party names, and with search by subject without and with the citation prior. Existence checks remove most fabricated cases; only search by subject, best with the prior, makes the models name the judgments that decide the questions.

II. RELATED WORK

Legal hallucination. Dahl et al. asked LLMs verifiable questions about United States federal cases and found hallucination rates between 58% (ChatGPT 4) and 88% (Llama 2) [13]. Magesh et al. found that commercial retrieval-augmented legal research tools still hallucinate on 17–33% of queries [14]. Liu et al. collected over 1,000 court filings containing fabricated citations and benchmarked automatic detectors on 1,300 brief excerpts with injected errors [15]. LegalCiteBench evaluates 21 models on citation tasks built from 1,000 United States opinions and finds that even the strongest scores below 7/100 on citation retrieval and completion [16]; Verma shows that models catch substituted cases far more reliably than wrong pinpoint pages [17]. All of these studies concern United States law.

Citation tools. eyecite extracts and normalises United States legal citations [18], and CourtListener’s citation lookup API matches them against its database to flag invalid ones [5]. Our verifier plays the same role for the Supreme Court of India, whose citation formats (neutral citations, S.C.R. supplementary volumes) and party-name conventions differ.

Citation networks and case retrieval. Fowler et al. built the citation network of all majority opinions of the United States Supreme Court and derived measures of the legal importance of precedents [6]. For India, Minocha et al. studied a citation network of judgments that invoke Articles 264–300 of the Constitution [19]. Prior-case retrieval asks for

the earlier judgments a case should cite. The Indian IL-PCR benchmark takes its relevance judgments from the citations in the judgments themselves [20], as citation recommendation does in general [21]. Retrieval-augmented generation grounds a model’s answer in retrieved documents [22]. We build the citation network for the whole reported Supreme Court, use it both as a ranking signal and as relevance labels, and measure what retrieval changes in the authorities that models cite.

Agents with citation checks. Recent legal agents retrieve sources before answering and audit their own citations: L-MARS measures citation faithfulness claim by claim [23], an agentic retrieval system answers 95.3% of 300 multiple-choice questions from the Korean bar examinations [24], Maat grounds competition-law answers in official decisions [25], and Parthenon wraps agents in deterministic checks for sources, dates and numbers [26]. A recent survey reviews this line of work [27]. Our verifier can serve as such a check for Indian law.

Indian legal NLP resources. ILDC provides 35k Supreme Court judgments for judgment prediction [28], IL-TUR benchmarks Indian legal text understanding [29], and NyayaRAG studies retrieval-augmented judgment prediction [30]. Juvekar et al. evaluate frontier models on Indian legal examinations and name citation discipline as a recurring failure [11], and Hemrajani’s survey experiment finds that LLMs frequently fabricate authorities in Indian legal research tasks [10]. Closest to our corpus, open-india-law releases extracted text and embeddings for 12.8 million judgments from the same AWS archive and treats a citation graph as out of scope [12]. We add a documented audit of the archive’s metadata, a citation verifier, an open citation graph of the Supreme Court and a benchmark of how models cite.

III. AN OPEN INDEX OF INDIAN JUDGMENTS

A. Source and loading

The AWS Open Data registry hosts two datasets published under CC-BY-4.0 by Dattam Labs: judgments of the Supreme Court from 1950 [8] and of 25 High Courts [9], each with structured metadata in Parquet and the judgments as PDFs. We load only the metadata (no text and no PDFs) into one PostgreSQL 17 table, reading the Parquet files directly from S3 (for the large High Court files, only the needed columns). The result holds 19,319,963 records: 38,366 Supreme Court judgments and 19,281,597 High Court judgments and orders from the 25 High Courts (26 courts in all), with decision years from 1950 to 2026. Each record stores a link to the original PDF in the archive, never the PDF itself; the text of the Supreme Court judgments, used in §V, is extracted separately.

B. Audit and repair

High Court metadata reaches the archive through two paths. Records from the “mobile” path store only a bare file name where a bench-qualified path is needed, so the PDF link cannot be formed. The archive’s documentation notes that mobile records may lack a PDF-existence flag and recommends the

TABLE I
HIGH COURT RECORDS FROM THE ARCHIVE’S MOBILE PATH, AND THE
LINKS RESTORED BY RECONCILIATION AGAINST THE ARCHIVE’S FILE
LISTING.

High Court	Records	Mobile	Linked after repair
Bombay	2,446,330	643,216	537,441 (83.6%)
Allahabad	2,300,821	605,598	584,934 (96.6%)
Madhya Pradesh	693,219	42,695	5 (<0.1%)
Himachal Pradesh	292,894	7	7 (100.0%)
All four	5,733,264	1,291,516	1,122,387 (86.9%)

case number, decision date and order number as the record identity [31], but it does not quantify the problem. Table I does. 1,291,516 records in four High Courts (6.7% of the corpus) are affected, 97.8% of them decided in 2024–2026.

The file name is not a usable identity either: 61,125 file names occur in more than one bench of the same court, covering 123,131 records. In 88.3% of these the title and the decision date both differ, so the records are different orders. Keying on the file name, as our first load did, silently overwrites at least 62,006 records. The case number (CNR) differs in all of them, so the documented identity is safe.

We repair the mobile records by taking the bench from the S3 key of the metadata file and then reconciling every record against a listing of the archive’s actual PDF files. A record is linked only when its file exists. This restores 1,122,387 links (86.9%). The remaining 169,129 records have no matching PDF in the archive’s file listing; they include all but 5 of Madhya Pradesh’s 42,695 mobile records. After repair 99.1% of all records carry a link.

C. Validation

We checked links against the live archive with HTTP HEAD requests. All 300 randomly sampled restored links resolve. For 450 randomly sampled unrecoverable records we requested the expected file in the decision-year folder and the two neighbouring years; none exists. For the original, non-mobile links we drew a per-court random sample: all 454 resolve, across 24 High Courts, as do all 150 sampled Supreme Court links. Of 60 randomly sampled file names shared by two benches, all 60 pairs point to different files (different size and S3 ETag).

D. Supreme Court citation metadata

Every one of the 38,366 Supreme Court judgments carries an S.C.R. citation, 7,642 of them (19.9%) in a supplementary volume (e.g., [1995] SUPP. 2 S.C.R. 359), and 38,238 carry a neutral citation. The Supreme Court launched neutral citations in 2023 in the form 2023INSC1: year of pronouncement, INSC, serial number [32]. Three properties of the metadata matter for verification:

- 1) The year field of a Supreme Court record is the year of its S.C.R. volume, not of the decision; they differ for 5,179 judgments (13.5%). We use the decision date throughout.

- 2) 96 neutral citations are attached to more than one record (216 records), for example a 2018 referral order that carries the neutral citation of the 2020 judgment in the same matter.
- 3) For each year from 1950 to 2013 except 1961, 1994 and 1997, the index contains every neutral-citation number from 1 to the year’s maximum. This matches how the numbers were issued: for 1950–2013 the Court assigned neutral citations retrospectively to the judgments reported in the S.C.R. [33]. From 2014 the Court numbers judgments and orders whether or not they are reported, and the index, which holds reported judgments, contains between 17% and 79% of each year’s numbers. A neutral citation missing from the index is therefore unassigned only for the 61 complete years.

E. Serving

A small HTTP service exposes keyword search, filters and exact lookups for CNR, neutral and S.C.R. citations. On a laptop (Intel Core i7-13700HX, 32 GB RAM, NVMe SSD) with default PostgreSQL settings, exact lookups answer in a median of 2.0 ms over HTTP; keyword queries are slower and depend on phrase frequency (§IX).

IV. CITATION VERIFIER

The verifier takes a case name, a year and a citation string, as a model produced them, and decides whether they denote a real Supreme Court judgment in the index.

Citations. Neutral citations are parsed with the pattern *year INSC number*. S.C.R. citations are parsed in the bracketed official form and the common variants (1978) 2 SCR 621 and 1995 (3) SCR 1; a citation without a volume number (common for supplementary volumes, e.g. [1973] Supp. S.C.R. 1) matches any volume of the same series, year and page. Citations to other law reports (SCC, AIR and others) cannot be resolved against the index and are left to the name match.

Names. A case name and a judgment’s party names are split into words and initials. Words are lower-cased, stop-words such as *the*, *v.* and *ors.* are dropped, and abbreviations are expanded (*C.B.I.*, *A.D.M.*, and state codes such as *U.P.* when they follow “of”); single letters written as initials, as in *K.S. Puttaswamy*, are kept apart. Each word t has weight $w(t) = \log(N/\text{df}(t))$ over the N Supreme Court judgments [34]; a word absent from the index gets weight $\log N$. Two words match if they are equal or, for words of five or more letters, if their similarity ratio [35] is at least 0.85, which accepts transliteration variants such as *Keshavananda/Kesavananda* but not *Patil/Patel*. Once a distinctive word (one found in at most 100 judgments) has matched, a rare given name may also match an initial in the judgment’s title, as *Shivkant* matches the “S.” of “S. S. Shukla”. With M_a and M_b the matched words of the name a and of the judgment b , let $p = W(M_a)/W(a)$ and $r = W(M_b)/W(b)$, where W sums word weights. The name score is the harmonic mean of p and r , set to zero if $p < 0.8$, so that an unmatched given name (*Kavita Devi* against *Ram*

Kumari Devi) cannot be outweighed by common words such as *State* or *Uttar Pradesh*, and set to zero if the initials in the cited name do not appear in the judgment’s title, either as initials or, after a distinctive match, as the first letters of names the title spells out (*R.D.* for *Ramana Dayaram*). Candidates are judgments whose decision year or S.C.R. volume year lies within one year of the stated year.

Labels. With threshold $\tau = 0.5$: *VERIFIED* if some candidate’s name score is at least τ and a neutral or S.C.R. citation, if given, points to that judgment or to one with matching parties; *WRONG_CITATION* if the parties match but the citation points elsewhere or nowhere; *MISATTRIBUTED* if no candidate matches but the citation resolves to a different judgment; *WRONG_YEAR* if neither resolves but the name identifies exactly one judgment (or several orders in one matter) elsewhere in the corpus, with a score of at least 0.8 and a word found in at most 15 judgments, i.e. a real case cited with the wrong year; *NOT_FOUND* otherwise. *VERIFIED* establishes that a judgment with those parties exists near that year, not that it supports the proposition for which it was cited.

Development. We developed these rules and thresholds on the T4 answers of three models, reviewing by hand a development sample of 56 decisions and every label that changed between versions of the rules. We then froze the verifier, recording hashes of its code, before running T5, scoring, and drawing the held-out audit reported in §VIII-D. One OCR error in the title of a benchmark judgment (*T.M.A. Pal for Pai*) is corrected when the index is loaded.

V. JUDGMENT TEXT AND CITATION GRAPH

A. Text and headnotes

We downloaded the PDF of every Supreme Court judgment in the archive and extracted its text layer with PyMuPDF. The reports are scanned pages with an embedded OCR layer, so no further OCR is needed; we keep only the text. 38,357 of the 38,366 judgments were extracted; for 9 the archive returned an error or an HTML page instead of the PDF. Every judgment in the S.C.R. opens with the reporter’s headnote: the parties, date and bench, then a summary of the questions decided and the holdings, then the judgment proper, usually introduced by a jurisdiction line such as “CIVIL APPELLATE JURISDICTION”. We cut the headnote at that line, keep at most 3,000 characters, and remove running page heads and the margin letters A–H of the printed report.

B. Extracting and resolving citations

We find every reporter citation in the text: S.C.R. and neutral citations, SCC citations, and AIR citations of the Supreme Court (AIR citations of other courts are discarded). Each distinct pair of citing judgment and citation is resolved in three tiers:

- 1) *Number*: an S.C.R. or neutral citation is looked up exactly in the index.
- 2) *Parallel number*: an SCC or AIR citation written directly beside an S.C.R. or neutral citation, as in (2012) 1

TABLE II
CITATIONS OF OTHER REPORTED JUDGMENTS, BY DECADE OF THE CITING JUDGMENT, AND HOW THEY WERE RESOLVED.

Citing decade	Citations	Resolved (%)	Resolved by		
			Number	Parallel	Name
1950s	953	87.6	821	0	14
1960s	3,259	84.0	2,669	6	63
1970s	5,001	75.9	3,561	18	219
1980s	6,467	77.5	4,160	43	806
1990s	15,667	63.8	4,958	361	4,676
2000s	34,810	55.0	2,957	375	15,830
2010s	70,340	78.5	23,027	17,174	15,026
2020s	67,418	84.0	26,370	22,036	8,212

SCC 40 : [2011] 13 SCR 309, takes that number’s judgment.

- 3) *Name*: otherwise the case name written before the citation and the citation’s year go through the frozen verifier of §IV, and the pair is resolved only if exactly one judgment matches.

If a number and the case name beside it point to different judgments, the pair is dropped. A citation of the judgment itself (its own report) is ignored. We developed the extractor on a pilot sample of 502 judgments and then ran it once over the whole corpus; the manual check in §V-C uses only judgments outside the pilot.

C. Size and accuracy

Across the 38,357 judgments the extractor found 203,915 distinct pairs of a citing judgment and a reporter citation of another judgment. 153,382 of them (75.2%) were resolved: 68,523 by number, 40,013 through a parallel number and 44,846 by case name and year. 749 pairs whose number and name disagreed were dropped. The resolved pairs form 109,370 distinct links between 22,483 citing and 21,518 cited judgments, so 56.1% of the Supreme Court’s reported judgments are cited by at least one later reported judgment. Resolution depends on the era of the citing judgment (Table II). Until the 1980s judgments cite mostly the S.C.R. and are resolved by number. From the 1990s the commercial SCC is the most cited report. Parallel S.C.R. citations, which make recent judgments easy to resolve, become common only in the 2010s, so the 2000s are the hardest decade.

Accuracy. Where a citation carries an S.C.R. or neutral number and the case name beside it can also be resolved, the name and year alone reach the number’s judgment in 98.4% of 46,091 cases. This is a label-free estimate of the precision of name-only links. Only 53 links (0.05%) point to a judgment decided after the citing one, which is impossible. In a sample of them the causes were wrong decision dates in the metadata, page numbers of a neighbouring judgment in the same report volume, and truncated generic names (*Others v. The State of Assam*) matched to the wrong judgment. Because name matching is restricted to the cited year, most wrong links cannot point forward, so this check is a sanity check, not an error bound. We also checked by hand 60 randomly drawn

name-only links from judgments outside the pilot, against the citation context and the index record: all 60 were correct, including names misspelt in the citing text, such as *A.N. Shegal* for *A.N. Sehgal*. Of 40 randomly drawn SCC and AIR citations that stayed unresolved, 15 cite judgments that are not in the index at all, because they were reported in the SCC or AIR but not in the S.C.R. The other 25 could have been resolved: 18 were lost to OCR errors or truncated names in the citing text, 3 to errors in the source (a wrong year in the citation, a misspelt index title), 2 to names shared by several judgments, and 2 to limitations of the frozen name parser.

Face validity. The most cited judgments are *Maneka Gandhi v. Union of India* (261 citing judgments), *Kesavananda Bharati* (170), *Sharad Birdhi Chand Sarda v. State of Maharashtra* (155), *Ramana Dayaram Shetty v. International Airport Authority* (150) and *Shivaji Sahebrao Bobade v. State of Maharashtra* (141): leading authorities on personal liberty, the basic structure, circumstantial evidence, State action and criminal appeals. Of the 51 landmark judgments that we chose by hand for the benchmark before the graph existed, 50 are cited at least once, and their median in-degree is higher than that of 98.9% of all judgments.

VI. FINDING AUTHORITIES BY SUBJECT

A. Retrievers

We index the title and headnote of every judgment in two ways. *BM25* [36] scores unigrams and bigrams (English stop words removed). *Dense* retrieval embeds each headnote with the open multilingual model bge-m3 [37], run locally, and scores by cosine similarity with the embedded query. The *hybrid* retriever takes the union of the top 300 judgments of each and adds their scores after normalising each to zero mean and unit variance over the candidates. A *citation prior* adds the normalised $\log(1+c)$, where c is the number of judgments in the graph that cite the candidate. Every combination is a weighted sum of these three scores; the weights are chosen on a development split and never on the evaluation data.

B. Evaluation without human labels

The graph supplies relevance judgments without annotation: the judgments a decision cites are the authorities its bench considered, the standard setting of citation recommendation [21] and of Indian prior-case retrieval [20]. For each held-out judgment that cites at least 5 earlier judgments, the query is its headnote with every reporter citation, every “X v. Y” case name and every “...’s case” reference removed, so the query states the legal issues without naming the precedents. The targets are the judgments it cites. Only judgments decided before it are candidates, and the citation prior counts only citations made before it, so no information from the future is used. We choose weights on 300 development judgments and report recall@10, recall@50, the share of queries with at least one target in the top 10 (hit@10) and mean reciprocal rank on 1,000 other test judgments, with bootstrap 95% intervals. We also score the 40 T4 questions against their reference judgments, with the weights from the development split.

TABLE III

CITATION RECOMMENDATION ON 1,000 HELD-OUT JUDGMENTS: GIVEN THE HEADNOTE WITHOUT CITATIONS AND CASE NAMES, RETRIEVE THE EARLIER JUDGMENTS IT CITES. $R@k$ = SHARE OF CITED JUDGMENTS IN THE TOP k ; HIT@10 = SHARE OF QUERIES WITH AT LEAST ONE IN THE TOP 10; MRR = MEAN RECIPROCAL RANK OF THE FIRST. WEIGHTS CHOSEN ON 300 OTHER JUDGMENTS.

Method	R@10	R@50	Hit@10	MRR
Most cited so far	0.013	0.039	0.111	0.056
BM25	0.151	0.321	0.662	0.390
Dense (bge-m3)	0.136	0.286	0.614	0.362
BM25 + dense	0.167	0.345	0.689	0.420
BM25 + prior	0.182	0.381	0.740	0.455
Dense + prior	0.159	0.323	0.686	0.426
BM25 + dense + prior	0.204	0.399	0.774	0.496

TABLE IV

RETRIEVAL FOR THE 40 T4 QUESTIONS: QUESTIONS WHOSE TOP- k RESULTS CONTAIN A REFERENCE JUDGMENT, AND THE SHARE OF REFERENCE JUDGMENTS IN THE TOP 10 ($R@10$). WEIGHTS FROM THE RECOMMENDATION DEVELOPMENT SPLIT.

Retriever	Top 3	Top 5	Top 10	R@10
Most cited	2	2	2	0.04
BM25	25	28	30	0.63
Dense	29	31	35	0.76
BM25 + dense	28	30	36	0.77
BM25 + dense + prior	33	35	37	0.84

C. Results

Citation recommendation. 7,047 judgments cite at least 5 earlier judgments that the graph resolves. On the 1,000 test judgments, which cite 11.2 of them on average, the hybrid retriever with the citation prior places 20.4% of the cited judgments in its top 10, and at least one of them for 77.4% of the queries (Table III). The prior helps every retriever. On the same queries it raises the recall@10 of the hybrid retriever by +0.036 (paired bootstrap 95% CI +0.030 to +0.043; better on 350 queries, worse on 99), a relative gain of 22%. Popularity alone is useless (0.013). The prior only reorders judgments that already match the subject, and among those, the ones the Court has cited often are more likely to be cited again. The query scrubbing leaves some names behind: 3.1% of the cited judgments share a rare title word (one found in at most 50 titles) with the query. Scoring only the other targets changes little (0.194 with the prior, 0.159 without).

The 40 legal questions. Searched with the question text, the hybrid retriever with the prior returns a judgment that decided the question among its top three results for 33 of the 40 questions, and among its top ten for 37 (Table IV). No model named such a judgment from memory for more than 24 questions. Every judgment the retriever returns exists, with its correct citations, by construction. The questions were written around landmark judgments, which are cited often, so this set favours the prior. The recommendation experiment has no such bias.

VII. HOW LANGUAGE MODELS CITE: SETUP

A. Items

The benchmark uses 101 Supreme Court judgments: 51 landmark judgments that we identified by name and confirmed individually in the index, and 50 judgments sampled at random (seeded) across eight periods from 1950 to 2023. Landmark candidates were dropped when the index record was ambiguous or erroneous (e.g., a shared neutral citation). All benchmark judgments have a unique neutral citation whose year equals the decision year.

- **T1, real citation** $\rightarrow$ **case** (202 items): each judgment is asked for once by its S.C.R. citation and once by its neutral citation: *Which judgment of the Supreme Court of India is reported as ... ?* The answer is correct if the named parties match the true judgment (name score $\geq \tau$).
- **T2, fabricated citation** (60 items): the same question for citations that cannot exist: 30 neutral citations in complete years with a number between $2m + 100$ and $3m + 100$, where m is the year’s last issued number, and 30 S.C.R. citations to a regular volume at least three above the year’s last. The correct response is to decline.
- **T3, case** $\rightarrow$ **citation** (101 items): given the parties and decision date, the model gives the S.C.R. and neutral citations; each is scored by exact match (with the volume rule above).
- **T4, authorities for a question** (40 items): the model lists up to three Supreme Court judgments that are leading authorities on a legal question (e.g., whether the right to privacy is a fundamental right), each with year and citation; every case is labelled by the verifier.
- **T5, revision after feedback**: for each T4 answer with an unverified case, the model is told which cases the index could not confirm and asked to revise its list.

For T4 we also record reference answers: for each question, the Supreme Court judgments that decided it, one to four per question (70 index records, since a matter decided in several orders contributes each order), including earlier decisions later overruled, such as *Suresh Kumar Koushal* for Section 377. We compiled them from the questions after the closed-book runs and before the search runs (§VII-C), and recorded their hash. A question is answered with a reference judgment if one of the listed cases matches it under the verifier’s name and year rules. This is a lower bound on relevance, since the models also cite other pertinent judgments.

B. Models and inference

We evaluate open-weight models that run on an 8 GB laptop GPU (NVIDIA RTX 4060) through Ollama [38], all in 4-bit (Q4_K_M) quantisation (Table V). Every request uses the same system prompt, which asks the model to state only facts it is confident about and to say when it does not know; temperature 0; a fixed seed; and a JSON schema that constrains the output format, so answers can be parsed without heuristics. Thinking is disabled for the Qwen3 model; as an ablation we also run it with thinking enabled on T1, T2 and T4, allowing

TABLE V
MODELS EVALUATED (OLLAMA BUILDS, 4-BIT Q4_K_M).

Model	Ollama tag	Params	Digest
Llama 3.1 8B [39]	llama3.1:8b	8.0B	46e0c10c
Qwen3 8B [40]	qwen3:8b	8.2B	500a1f06
Qwen3 8B think [40]	qwen3:8b	8.2B	500a1f06
Gemma 3 4B [41]	gemma3:4b	4.3B	a2af6cc3
Qwen2.5-VL 7B [42]	qwen2.5vl:7b	8.3B	5ced39df

3,000 additional output tokens for the reasoning and an 8,192-token context.

C. Search conditions

In a *name search* condition the models answer the same T1–T4 items with access to the index metadata. The Supreme Court records hold party names, dates and citations but no text, so this search cannot retrieve judgments by subject: it can confirm a judgment the model recalls and supply its citations. The model has two tools. *Search* returns the 6 judgments whose party names best match the query, ranked by the IDF-weighted name score of §IV over judgments that share one of the query’s three rarest words or a spelling variant of it. *Lookup* returns the judgment with a given neutral or S.C.R. citation, or reports that the index has none. The task prompt is followed by a description of the tools, including what the index holds. The model replies with one action at a time in JSON, receives each result as the next message, and after at most 5 actions is asked for its final answer in the task’s own schema, which is scored exactly like the closed-book answer. We use this JSON protocol rather than native tool calling because Gemma 3 has none in Ollama. The context is 8,192 tokens; all other settings are unchanged. We tried the protocol on four items outside the benchmark, froze it with the reference answers by recording hashes, and then ran it once.

In two *topic search* conditions we repeat T4 with the retriever of §VI as the search tool. The protocol is unchanged, but *search* now matches the title and headnote, so the model can search by subject, and each result shows the parties, date, citations and the first 280 characters of the headnote. We use the hybrid retriever, once without and once with the citation prior (weights from the recommendation development split). We checked the protocol on one question outside the benchmark and froze the retrieval code by hash before these runs.

D. Metrics

For T1 and T2 we report the share of items answered correctly, answered wrongly, and declined; invalid outputs (e.g., a degenerate repetition that hits the length limit) are counted separately. For T4 and T5 we report the share of cited cases in each verifier label, and the number of questions whose list includes a reference judgment. Where we give 95% intervals they are Wilson intervals [43]; for T4 they treat cited cases as independent, although cases cited for the same question are related, so they understate the uncertainty.

TABLE VI

IDENTIFYING A JUDGMENT FROM ITS CITATION. T1: 202 REAL CITATIONS (101 JUDGMENTS, EACH BY S.C.R. AND NEUTRAL CITATION). T2: 60 CITATIONS THAT CANNOT EXIST. CORR. = NAMED THE RIGHT JUDGMENT; NAMED = NAMED ANY JUDGMENT FOR A FABRICATED CITATION; DECL. = DECLINED; INV. = INVALID OUTPUT.

Model	T1: real citation (%)				T2: fabricated (%)		
	Corr.	Wrong	Decl.	Inv.	Named	Decl.	Inv.
Llama 3.1 8B	0.5	99.0	0.0	0.5	98.3	0.0	1.7
Qwen3 8B	0.0	0.0	100.0	0.0	0.0	100.0	0.0
Qwen3 8B think	1.0	14.4	83.2	1.5	10.0	88.3	1.7
Gemma 3 4B	0.0	100.0	0.0	0.0	100.0	0.0	0.0
Qwen2.5-VL 7B	0.0	0.0	100.0	0.0	0.0	100.0	0.0

VIII. HOW LANGUAGE MODELS CITE: RESULTS

A. Identifying citations (T1, T2)

Table VI shows two opposite behaviours. Llama 3.1 and Gemma 3 answered almost every question and were almost always wrong. Llama identified 1 of the 202 real citations (*Kesavananda Bharati*, by its S.C.R. citation); Gemma identified none. Both named a judgment for nearly every fabricated citation: 59 of 60 for Llama (98.3%, 95% CI 91.1–99.7%; one output was invalid) and 60 of 60 for Gemma (100.0%, 95% CI 94.0–100.0%). Their wrong answers were not near misses: the closest any wrong T1 answer came to the true judgment was a name score of 0.36 (Llama) and 0.34 (Gemma), well below the match threshold of 0.5. The answers also repeated: across T1 and T2, Llama gave *State of Punjab v. Davinder Pal Singh Bhullar* as the answer 44 times, and Gemma gave *State of Rajasthan v. Ram Manohar Arora* 80 times. The two Qwen models took the opposite course and declined all 262 questions. Under our system prompt, declining is the correct response to T2, but it forfeits T1 entirely. With thinking disabled, no model identified more than 1 of the 202 real citations.

B. Supplying citations (T3)

When given the parties and the date of a judgment and asked for its citations, the two Qwen models, which had declined every T1 question, supplied a neutral citation for most judgments, and almost all were wrong (Table VII). Across the four models only 3 of the 808 requested citations were correct. Qwen2.5-VL returned a neutral citation for 91 of 101 judgments and none was correct; 50 of them carry a number above the year’s last issued neutral citation in a year the index covers completely, so they cannot exist. Qwen3 returned wrong neutral citations for 81 judgments, 52 of which belong to other judgments, and wrong S.C.R. citations for 84, 50 of which point to a page where no reported judgment begins. Llama answered the request for a neutral citation with an AIR citation 97 times.

C. Authorities for legal questions (T4)

Asked for up to three leading Supreme Court authorities on each of 40 questions, the models named between 102 and 117 cases each, counting a case repeated within one

TABLE VII

SUPPLYING THE CITATIONS OF 101 KNOWN JUDGMENTS, GIVEN PARTIES AND DECISION DATE. CORR. = EXACT MATCH; WRONG = ANY OTHER NON-EMPTY ANSWER; EMPTY = NO CITATION GIVEN.

Model	S.C.R. citation (%)			Neutral citation (%)		
	Corr.	Wrong	Empty	Corr.	Wrong	Empty
Llama 3.1 8B	2.0	96.0	2.0	0.0	98.0	2.0
Qwen3 8B	0.0	83.2	16.8	1.0	80.2	18.8
Qwen3 8B think	–	–	–	–	–	–
Gemma 3 4B	0.0	100.0	0.0	0.0	100.0	0.0
Qwen2.5-VL 7B	0.0	11.9	88.1	0.0	90.1	9.9

TABLE VIII

T4: EVERY CASE CITED FOR 40 LEGAL QUESTIONS, BY VERIFIER LABEL. VER. = VERIFIED (WITH WILSON 95% INTERVAL), W.Yr. = WRONG_YEAR, W.Cit. = WRONG_CITATION, MISATT. = MISATTRIBUTED, N.F. = NOT_FOUND.

Model	Cases	Share of cited cases (%)					
		Ver.	95% CI	W.Yr.	W.Cit.	Misatt.	N.F.
Llama 3.1 8B	114	61.4	52–70	1.8	0.0	0.0	36.8
Qwen3 8B	117	50.4	42–59	7.7	0.0	0.0	41.9
Qwen3 8B think	94	35.1	26–45	4.3	0.0	0.0	60.6
Gemma 3 4B	117	23.9	17–32	6.8	0.0	0.0	69.2
Qwen2.5-VL 7B	102	22.5	16–32	0.0	0.0	11.8	65.7

answer once (Table VIII). Only 23–61% of these could be matched to a Supreme Court judgment with the named parties within a year of the stated date; between 36.8% (Llama) and 69.2% (Gemma) were NOT_FOUND. The models cited almost exclusively the commercial SCC and AIR reports and SCC OnLine, which the index cannot resolve, so these labels rest on the party names and year. Only Qwen2.5-VL used the S.C.R. form to any extent (13 citations, 12.7% of its total). All 12 of these that resolved in the index pointed to a different judgment (MISATTRIBUTED): for example, it attributed 1970 (2) SCR 1, which is *Sitabai v. Ram Chandra*, to an invented *State of Maharashtra v. Keshav Bapuji Patil*. WRONG_YEAR cases are real judgments cited with a wrong year, such as *R. Rajagopal v. State of Tamil Nadu* (1994) cited as 1969 and 1985.

D. Verifier accuracy

We built the verifier on a development sample of T4 answers and then froze it. To measure it we drew a fresh, stratified random sample of 95 T4 decisions from the four models, excluding every case on the development sheet, and judged each by hand against the index and our knowledge of the case law. The verifier was right in 92 of 95 (96.8%). 39 of 40 VERIFIED labels were correct; the error matched *Narendra Kumar v. Union of India* to *State Bank of India v. Narendra Kumar Pandey*. 38 of 40 NOT_FOUND labels were correct; both misses were real judgments whose titles contain OCR errors in the source (*Machhi Singh* spelled “Machhl”, *Baliya* spelled “Baliay”). All 11 WRONG_YEAR and 4 MISATTRIBUTED labels were correct.

TABLE IX

T5: THE T4 ANSWERS THAT CONTAINED AN UNVERIFIED CASE (Q), BEFORE AND AFTER THE MODEL REVISED THEM WITH THE VERIFIER’S FEEDBACK.

Model	Q	Cases		Verified (%)		Not found	
		Before	After	Before	After	Before	After
Llama 3.1 8B	30	87	48	49.4	87.5	42	4
Qwen3 8B	29	84	27	30.9	92.6	49	1
Gemma 3 4B	37	111	40	19.8	52.5	81	16
Qwen2.5-VL 7B	35	87	39	9.2	20.5	67	31

The audit also shows what NOT_FOUND means. Of the 40 audited cases, 27 named parties that match no Supreme Court judgment in any year, 9 used names so common (*Rajesh Kumar v. State of Bihar*) that same-named judgments exist only in other years, 1 was a real judgment cited with both a wrong year and an altered name, 1 was not a judgment at all (a committee report), and 2 were the verifier misses above.

E. Revision after verifier feedback (T5)

For the 131 T4 answers that contained at least one unverified case, we told the model what the check had found for each case (for example, that no Supreme Court judgment with those parties exists near the stated year) and asked it to revise its list. Table IX shows the result. The models dropped the flagged cases: across the four models the number of NOT_FOUND cases fell from 239 to 52, and the share of verified cases rose for every model. The gain came entirely from removal. 147 of the 154 cases in the revised lists were repeated from the first answers; only Gemma named new cases (7 in all), and none of them could be verified. The number of verified cases fell slightly, from 99 to 96. Qwen3 returned an empty list for 7 questions, and Qwen2.5-VL’s revised lists still contained 31 cases the index could not find, all repeated from its first answers. Feedback from an existence check thus makes an answer shorter and safer, but not more complete.

F. Effect of reasoning

With thinking enabled, Qwen3 wrote a median of 2,631 characters of reasoning per item and took a median of 16.0 s instead of 0.5 s (Table VI, row “Qwen3 8B think”; T3 was not run). It stopped declining every question, and the additional answers were mostly wrong. On T1 it identified 2 real citations correctly and 29 wrongly, declining 168; on T2 it named a judgment for 6 of the 60 fabricated citations (10.0%, 95% CI 4.7–20.2%), sometimes a famous one: it identified the non-existent [1992] 7 S.C.R. 142 as *Indra Sawhney v. Union of India* and [1997] 7 S.C.R. 501 as *Vishaka v. Rajasthan*. On T4 the share of verified cases fell from 50.4% to 35.1% and the share of NOT_FOUND cases rose from 41.9% to 60.6%; 5 of the 40 answers were invalid because the reasoning exhausted the token budget (9 items in all). At this scale, reasoning made the model more willing to answer without making it more accurate.

TABLE X

CITATION TASKS WITH SEARCH AND LOOKUP (COMPARE TABLES VI AND VII). T1: 202 REAL CITATIONS; T2: 60 FABRICATED CITATIONS; T3: SHARE OF THE 101 JUDGMENTS WHOSE S.C.R. OR NEUTRAL CITATION WAS SUPPLIED EXACTLY.

Model	T1 (%)			T2 (%)		T3 corr. (%)	
	Corr.	Wrong	Decl.	Named	Decl.	S.C.R.	Neutr.
Llama 3.1 8B	99.0	1.0	0.0	1.7	98.3	73.3	73.3
Qwen3 8B	97.5	2.5	0.0	0.0	100.0	100.0	100.0
Gemma 3 4B	49.5	49.0	1.5	51.7	48.3	99.0	99.0
Qwen2.5-VL 7B	86.6	0.0	13.4	0.0	100.0	100.0	100.0

G. Searching by name

With access to the index the citation tasks became easy for most models (Table X). The models identified between 100 and 200 of the 202 real citations, against at most 1 without search, and three of them supplied the correct S.C.R. and neutral citations for nearly every T3 judgment. Llama and Qwen3 looked up the given citation in every T1 and T2 item. After an empty lookup Llama named a judgment for only 1 fabricated citation, against 59 without search. Two weaknesses remain. Gemma looked up the given citation in only 102 of the 202 T1 items. In the others it searched for a case name it had guessed, never saw the right judgment, and answered from those results or declined; it gave 99 wrong answers in all. For 31 of the 60 fabricated citations it still named a judgment, 10 of them real judgments that its own searches for invented names had returned. Qwen2.5-VL declined 27 T1 items although the lookup had returned the right judgment. Some answers scored wrong do name the right judgment, with a small copying error that the strict name matcher rejects (*Irrc* for *IRRC*, *Simplicity* for *Simplex*, or a title copied with an empty respondent, “... v.”, whose “v” the frozen parser reads as an initial). This applies to 2 of Llama’s 2 and 5 of Qwen3’s 5 wrong T1 answers. We count them as wrong, as scored.

For legal questions the effect differs (Table XI, name search). Name search almost eliminated cases that the index cannot find: across the four models, NOT_FOUND cases fell from 239 to 8, and verified cases rose from 180 to 351. But fewer answers contained a judgment that decided the question: 27 of the 160 model–question pairs, against 48 without search. Question by question, 24 pairs lost their reference judgment and 3 gained one (exact McNemar test, $p < 0.001$).

The cause is how the models searched. The tool description stated that the index holds no subject matter, yet Llama and Qwen2.5-VL mostly searched with the words of the question: 46 of Llama’s 76 searches and 60 of Qwen2.5-VL’s 77 consisted mainly of the question’s own words. They then listed the judgments whose party names happened to contain those words. On whether “procedure established by law” must be fair, just and reasonable, Llama had cited *Maneka Gandhi* without search; with search it listed *Fair Air Engineers v. N.K. Modi* and *Just Society v. Union of India*. Qwen2.5-VL listed *State of U.P. v. U.P. State Law Officers Association* and *Bajaj Auto v. Company Law Board*. These judgments exist, so the

verifier labels them VERIFIED. Yet none of the 134 listed judgments that came only from such searches is a reference judgment.

Qwen3 searched by party names (only 7 of its 78 searches consisted mainly of the question’s words) and kept nearly all its reference hits (13 questions without search, 12 with): search confirmed the judgments it recalled but added none it did not. Gemma also searched by names, but paired the citations it found with names from memory, producing 28 MISATTRIBUTED cases, such as *Jacob Mathew v. State of Kerala* with the citation of *Jacob Mathew v. State of Punjab*. Gemma and Qwen2.5-VL often wrote their whole answer into the *answer* action (26 and 21 of 40 T4 episodes), which was cut at the length limit; they were then asked for the final answer as in every other episode.

H. Searching by subject

Topic search changed the models’ answers to the legal questions more than any other condition (Table XI). Across the four models, answers contained a judgment that decided the question in 109 of the 160 model–question pairs with topic search and in 133 with topic search and the citation prior, against 48 from memory and 27 with name search. Compared question by question with the answers from memory, topic search with the prior gained a reference judgment in 96 pairs and lost one in 11 (exact McNemar test, $p < 0.001$). Adding the prior to topic search gained 27 and lost 3 ($p < 0.001$). The gain was largest for the model that knew least: Qwen2.5-VL, which named a reference judgment for 3 questions from memory, named one for 33. At the same time almost every listed case exists. 98.5% of the 463 cases listed with topic search and the prior are VERIFIED and 1 is NOT_FOUND, against 40.0% and 239 from memory. The models listed mostly judgments that the search had shown them: in every model and condition between 93.8% and 97.5% of the listed cases had appeared in the search results.

The two conditions show what each part contributes. Topic search alone returns real judgments on the subject, and the prior moves the leading authority among them. On workplace sexual harassment, Qwen2.5-VL named a non-existent *Sukanya v. Union of India* from memory. With topic search it listed *Medha Kotwal Lele v. Union of India* and *D.S. Grewal v. Vimmi Joshi*, which exist and concern the subject, and with the prior it listed *Vishaka*. On whether Article 21 rights can be suspended during an emergency, Llama’s topic search returned emergency-era detention cases such as *Makhan Singh v. State of Punjab*; with the prior it listed *A.D.M. Jabalpur*.

Listing the top three results of the retriever, with no model at all, answers 33 questions with the prior and 28 without (Table IV). The models answered 35, 33, 32 and 33 questions with the prior and between 24 and 29 without.

IX. DISCUSSION

Existence and authority are different checks. A verifier over an index of names and citations catches fabricated cases, and feedback from it or search over it removed almost all

of them. It did not help the models find the judgment that decides a question, and name search made that rarer, because the models searched with the question’s words and listed any judgment whose parties happened to share them. A real but irrelevant judgment passes an existence check. A system that shows only “verified” citations can therefore look more reliable while citing worse authorities. Finding the authority needs what the metadata lacks: a description of what each judgment decided, which the reporter’s headnote supplies, and a record of which judgments the Court itself relies on, which the citation graph supplies. With both, the same small models named a deciding judgment in 133 of 160 cases and almost never cited a case that does not exist.

The retriever does the work. Listing the retriever’s top three results answers about as many questions as the models that use it (33 against 32–35), and the models listed almost only what the search had shown them. For a system that must not invent authorities this is the desirable division of labour: the index guarantees existence, the retriever and the graph supply relevance, and the model at most phrases and orders the result. The citation prior helps because it encodes the Court’s own judgment of importance, as network measures of precedent do for the United States Supreme Court [6]. It favours old and frequently cited judgments, including judgments later overruled, such as *A.D.M. Jabalpur*, which a lawyer must cite as overruled. Popularity alone is useless (recall@10 0.013), so the prior can only reorder judgments that already match the subject.

Declining one question is not reliability. The two Qwen models declined every request to identify a judgment from its citation, which looks like calibrated caution. Yet asked, with the same system prompt, for the citation of a judgment they had just been told exists, they produced one for most judgments, and nearly all were wrong. Whether a small model admits ignorance therefore depends on how the question is framed, and a single abstention test says little about how it will behave in drafting, where it is asked to supply citations rather than check them.

Neutral citations invite invention. The Supreme Court began issuing neutral citations in 2023 and assigned them retrospectively to its reported judgments from 1950 [32], [33], so for most judgments they are recent additions that a model is unlikely to have seen in use. The models nevertheless produced them readily, many with numbers that were never issued. The same property that makes them easy to fabricate makes them easy to check: for the 61 years whose numbering the index holds completely, a single lookup settles whether a neutral citation exists.

What remains unchecked. Neither the verifier nor the retriever checks that a judgment supports the proposition it is cited for, which needs the judgment’s reasoning and is hard even for strong models [17]. The graph records that one judgment cites another, not whether it follows, distinguishes or overrules it. Citations to the SCC and AIR reports are resolved through parallel S.C.R. citations or by party names, never by their own numbers, and judgments reported only in

TABLE XI

T4 UNDER FIVE CONDITIONS: FROM MEMORY, AFTER VERIFIER FEEDBACK (T5 ANSWERS WHERE REVISED, T4 OTHERWISE), WITH NAME SEARCH OVER THE INDEX METADATA, AND WITH TOPIC SEARCH OVER HEADNOTES WITHOUT AND WITH THE CITATION PRIOR. REF. = QUESTIONS (OF 40) WHOSE LIST INCLUDES A JUDGMENT THAT DECIDED THE QUESTION; N.F. = LISTED CASES THE VERIFIER COULD NOT FIND.

Model	From memory		After feedback		Name search		Topic search		Topic + prior	
	Ref.	N.F.	Ref.	N.F.	Ref.	N.F.	Ref.	N.F.	Ref.	N.F.
Llama 3.1 8B	24	42	24	4	10	4	29	2	35	1
Qwen3 8B	13	49	13	1	12	1	24	0	33	0
Gemma 3 4B	8	81	8	16	5	1	27	0	32	0
Qwen2.5-VL 7B	3	67	3	31	0	2	29	0	33	0
All four	48	239	48	52	27	8	109	2	133	1

those reports are not in the index: 15 of the 40 unresolved citations we inspected fall in that gap. Recent neutral citations can only be confirmed, not refuted, because the index holds only reported judgments after 2013.

Limitations. We evaluated four open-weight models of roughly 4 to 8 billion parameters (Table V) in 4-bit quantisation, with one prompt and greedy decoding. Larger and commercial systems may know many more landmark citations, and different prompts may change the balance between answering and declining. The benchmark is small (101 judgments, 40 questions) and the landmark set is our own selection. The audit labels, the reference answers and the manual check of the graph come from a single annotator, working from the index, the judgment text and published sources, so we cannot report inter-annotator agreement. We compiled the reference answers after the closed-book runs. They list only judgments that decided each question, so they measure relevance only from below, and because they are landmarks they favour the citation prior. The recommendation experiment has neither problem, but it treats the judgments a decision cites as its relevant authorities, which is incomplete: a court cites some of the relevant judgments, not all of them. Each search condition used one tool description and one query method. The frozen name parser misreads a name that ends in “v.” with no respondent (two answers affected) and a party name that begins with the initial V. VERIFIED means that a judgment with the named parties exists near the stated year, not that it is the authority the model meant. The corpus is a snapshot of an archive that is updated periodically, and it has its own errors: 96 neutral citations are attached to more than one record, titles contain OCR errors, some judgments are filed under the wrong year, and 9 PDFs are missing.

Serving cost. Everything runs on one laptop with default PostgreSQL settings. We timed 129 queries to the index over HTTP, each run twice, directly after a database restart: exact lookups by CNR, neutral or S.C.R. citation took a median of 2.0 ms (at most 15 ms), and keyword queries for common phrases a median of 2,826 ms on first touch and 270 ms when repeated. The verifier checked a cited case in a median of 80 ms (95th percentile 1,992 ms). Embedding the headnotes ran at about 8 judgments per second on the laptop GPU, and a retrieval query, including embedding the query, took about 84 ms on average.

X. CONCLUSION

Indian courts now treat reliance on fabricated precedents as misconduct [3], so a model-drafted citation must be checked before it reaches a court. We built the checks from open data: an audited index of 19,319,963 judgments and orders, a verifier for Supreme Court citations whose labels were correct in 96.8% of a held-out audit, and an open citation graph of 109,370 links extracted from the text of every reported Supreme Court judgment, whose name-only links agree with citation numbers in 98.4% of checkable cases. Small open models answering from memory fabricate freely. An existence check, as feedback or as a search tool, removes the fabricated cases but not the lack of authority. Search by subject over the reporters’ headnotes, ranked with the citation counts of the graph, finds the judgments that decide a question: its top three results do so for 33 of 40 questions, and with it the same models named such a judgment in 133 of 160 cases while almost every case they listed exists. Checking existence is cheap and should run before any citation reaches a court. Finding authority needs text and the citation network. Checking that a judgment supports what it is cited for remains open.

ACKNOWLEDGMENT

An AI assistant (Claude, Anthropic) was used to write the software, run the experiments and analyses, produce the first-pass labels of the verifier audit, the manual check of the citation graph and the reference answers for T4, and draft the text. The author is responsible for the content.

DATA AND CODE AVAILABILITY

The index builder, repair scripts, verifier, citation extractor, citation graph (every resolved and unresolved citation with its resolution tier), headnote embeddings, retrieval code, benchmark items, prompts, raw model outputs and scoring code are available from the author. The underlying data are CC-BY-4.0 [8], [9].

REFERENCES

- [1] United States District Court for the Southern District of New York, “Mata v. Avianca, Inc., 678 F. Supp. 3d 443 (S.D.N.Y. 2023),” 2023, opinion and order on sanctions, June 22, 2023.

- [2] Bar & Bench, “Bombay High Court pulls up tax officer for citing fake, AI-generated case laws (KMG Wires Private Limited v. AO, NFAC),” <https://www.barandbench.com/news/litigation/bombay-high-court-pulls-up-tax-officer-for-citing-fake-ai-generated-case-laws>, Oct. 2025, published Oct. 30, 2025; accessed Sep. 29, 2026.
- [3] MediaNama, “SC says citing AI-generated fake case laws is misconduct (Gummadi Usha Rani v. Sure Mallikarjuna Rao, Supreme Court of India, order of Feb. 27, 2026),” <https://www.medianama.com/2026/03/223-supreme-court-ai-fake-case-laws-misconduct/>, Mar. 2026, accessed Sep. 29, 2026.
- [4] SCC Online, “Supreme Court sets aside Rs. 425.27-crore customs penalty order built on AI-generated fake citations (Vijay Ghanshyam Gadiya v. Union of India, 2026 SCC OnLine SC 1741, decided Sep. 2, 2026),” <https://www.scconline.com/blog/post/2026/09/03/supreme-court-sets-aside-%E2%82%B9425-crore-customs-penalty-over-ai-generated-fake-citations/>, Sep. 2026, accessed Sep. 29, 2026.
- [5] Free Law Project, “Combat hallucinations and look up citations with our new API,” <https://free.law/2024/04/16/citation-lookup-api/>, Apr. 2024, accessed Sep. 29, 2026.
- [6] J. H. Fowler, T. R. Johnson, J. F. S. II, S. Jeon, and P. J. Wahlbeck, “Network analysis and the law: Measuring the legal importance of precedents at the U.S. Supreme Court,” *Political Analysis*, vol. 15, no. 3, pp. 324–346, 2007.
- [7] Supreme Court of India, “Digital Supreme Court Reports (Digi-SCR),” <https://digisr.sci.gov.in/>, accessed Sep. 29, 2026.
- [8] Dattam Labs, “Indian Supreme Court judgments,” Registry of Open Data on AWS, <https://registry.opendata.aws/indian-supreme-court-judgments/>, s3 bucket indian-supreme-court-judgments (ap-south-1), CC-BY-4.0; accessed Sep. 29, 2026.
- [9] —, “Indian High Court judgments,” Registry of Open Data on AWS, <https://registry.opendata.aws/indian-high-court-judgments/>, s3 bucket indian-high-court-judgments (ap-south-1), CC-BY-4.0; accessed Sep. 29, 2026.
- [10] R. Hemrajani, “Evaluating the role of large language models in legal practice in India,” arXiv:2508.09713, 2025.
- [11] K. Juvekar, A. Bhattacharya, S. Khadloya, and U. Saxena, “Are LLMs court-ready? Evaluating frontier models on Indian legal reasoning,” arXiv:2510.17900, 2025.
- [12] Vaquill AI, “open-india-law: Open, structured Indian primary law,” <https://github.com/Vaquill-AI/open-india-law>, snapshot v2026.08.1; accessed Sep. 29, 2026.
- [13] M. Dahl, V. Magesh, M. Suzgun, and D. E. Ho, “Large legal fictions: Profiling legal hallucinations in large language models,” *Journal of Legal Analysis*, vol. 16, no. 1, pp. 64–93, 2024.
- [14] V. Magesh, F. Surani, M. Dahl, M. Suzgun, C. D. Manning, and D. E. Ho, “Hallucination-free? Assessing the reliability of leading AI legal research tools,” *Journal of Empirical Legal Studies*, vol. 22, no. 2, pp. 216–242, 2025.
- [15] P. Liu, D. Stambach, and P. Henderson, “Who checks the citations? Benchmarking legal hallucination detection,” arXiv:2606.21155, 2026.
- [16] S. Chen, H. Yin, and S. Zhou, “LegalCiteBench: Evaluating citation reliability in legal language models,” arXiv:2605.10186, 2026.
- [17] A. Verma, “Is this citation on point?” arXiv:2608.12571, 2026.
- [18] J. Cushman, M. Dahl, and M. Lissner, “eyecite: A tool for parsing legal citations,” *Journal of Open Source Software*, vol. 6, no. 66, p. 3617, 2021.
- [19] A. Minocha, N. Singh, and A. Srivastava, “Finding relevant Indian judgments using dispersion of citation network,” in *Proceedings of the 24th International Conference on World Wide Web (WWW Companion)*, 2015, pp. 1085–1088.
- [20] A. Joshi, A. Sharma, S. K. Tanikella, and A. Modi, “U-CREAT: Unsupervised case retrieval using events extraction,” in *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2023, pp. 13 899–13 915.
- [21] M. Färber and A. Jatowt, “Citation recommendation: approaches and datasets,” *International Journal on Digital Libraries*, vol. 21, pp. 375–405, 2020.
- [22] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel, S. Riedel, and D. Kiela, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 9459–9474.
- [23] B. Yuan and Z. Wang, “L-MARS: Legal multi-agent system with agentic search and citation-faithfulness audit,” in *Proceedings of the AIALaw Workshop at the 43rd International Conference on Machine Learning*, 2026, arXiv:2509.00761.
- [24] E. Y. Lim and J. Park, “Agentic RAG for legal question answering in civil law: Evidence from the Korean bar examination,” *IEEE Access*, 2026.
- [25] B. Mounir, F. Madkour, A. Abdelaziz, and A. Sami, “Maat: The agentic legal research assistant for competition protection,” arXiv:2605.27331, 2026.
- [26] H. Geng and L. Liu, “Parthenon Law: A self-evolving legal-agent framework,” arXiv:2606.04602, 2026.
- [27] S. Liu, R. Zhang, R. Ma, Y. Deng, L. Zhu, J. Li, Z. Li, Z. Shen, and M. Du, “LLM agents in law: Taxonomy, applications, and challenges,” in *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2026, pp. 15 768–15 792.
- [28] V. Malik, R. Sanjay, S. K. Nigam, K. Ghosh, S. K. Guha, A. Bhattacharya, and A. Modi, “ILDC for CJPE: Indian legal documents corpus for court judgment prediction and explanation,” in *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, 2021, pp. 4046–4062.
- [29] A. Joshi, S. Paul, A. Sharma, P. Goyal, S. Ghosh, and A. Modi, “IL-TUR: Benchmark for Indian legal text understanding and reasoning,” in *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Bangkok, Thailand, 2024, pp. 11 460–11 499.
- [30] S. K. Nigam, B. D. Patnaik, S. Mishra, A. V. Thomas, N. Shallum, K. Ghosh, and A. Bhattacharya, “NyayaRAG: Realistic legal judgment prediction with RAG under the Indian common law system,” arXiv:2508.00709 (AAACL-IJCNLP 2025), 2025.
- [31] Dattam Labs, “Indian High Court judgments: dataset documentation,” <https://github.com/vanga/indian-high-court-judgments/blob/main/opendata/docs/dataset.md>, accessed Sep. 25, 2026.
- [32] Supreme Court of India, “Notice: Launch and implementation of neutral citation system at Supreme Court of India in a phased manner,” Jul. 2023, notice dated July 6, 2023.
- [33] —, “Circular: Implementation of phase-II of the neutral citation system,” Sep. 2023, circular dated Sep. 27, 2023; covers reported judgments of 30.01.1950–31.12.2013 published in the Supreme Court Reports.
- [34] K. Spärck Jones, “A statistical interpretation of term specificity and its application in retrieval,” *Journal of Documentation*, vol. 28, no. 1, pp. 11–21, 1972.
- [35] Python Software Foundation, “difflib — helpers for computing deltas (SequenceMatcher),” Python 3.13 documentation, <https://docs.python.org/3/library/difflib.html>.
- [36] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: BM25 and beyond,” *Foundations and Trends in Information Retrieval*, vol. 3, no. 4, pp. 333–389, 2009.
- [37] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, “M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation,” arXiv:2402.03216, 2024.
- [38] Ollama, “Ollama,” <https://github.com/ollama/ollama>, version 0.34.4.
- [39] A. Grattafiori et al., “The Llama 3 herd of models,” arXiv:2407.21783, 2024.
- [40] Qwen Team, “Qwen3 technical report,” arXiv:2505.09388, 2025.
- [41] Gemma Team, “Gemma 3 technical report,” arXiv:2503.19786, 2025.
- [42] S. Bai et al., “Qwen2.5-VL technical report,” arXiv:2502.13923, 2025.
- [43] E. B. Wilson, “Probable inference, the law of succession, and statistical inference,” *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.